Problem Statements X AI Tech Ideathon

1.) **AI for Electoral Roll Transparency**

The Problem

Electoral rolls decide who gets to vote, yet the data behind them is huge, technical and hard for ordinary people to check. When large revisions add or delete millions of names, doubts and allegations follow, and the public has no easy way to separate fact from rumor.

Citizens can't tell whether their own name is safe. Journalists and observers can't audit the data at scale. Misinformation spreads faster than verified facts. Trust in elections suffers, whatever the truth turns out to be.

The Challenge

Build an AI-powered tool that makes voter-roll data **auditable, explainable and accessible**, without taking any political side.

What We Need

A. The Solution (at least one module, best teams will combine more)

1. **Anomaly detection:** Spot unusual patterns in roll data, such as abnormal deletions in one area, duplicate entries, impossible ages, or sharp changes between two versions of a roll.
2. **Plain-language explanations:** For every flag, say what was found, why it is unusual, and what innocent explanations exist.
3. **Citizen assistant:** A text or voice bot in Hindi, English and one more regional language that helps people check their registration and understand next steps.
4. **Claim checker:** Take a viral allegation and return what is verified, what is disputed, and what cannot be proven from public data.

B. The Content Video (one team member owns this)
A 45 to 90 second reel or ad that tells the story of your project. It should include:

- Behind-the-scenes moments of the team building during the hackathon
- A clear hook that states the problem in the first 5 seconds
- A short demo of the product
- AI used in making the video, for example AI voiceover, AI-generated visuals or animation, auto-captions, or AI editing tools.

Rules

- Use **synthetic or public data only**. No real voter data.
- Flag patterns for **human review**. Never declare fraud or name individuals, parties or institutions.
- Every output must show its **reasoning and confidence level**.
- Treat claims from all sides **equally**.

Deliverables

- A working prototype (app, bot or dashboard)
- A live demo on test data

2.) **AI for Event Reach and Registration**

The Problem

Hackathons, robotics olympiads and cultural competitions are run by small teams of student or community organizers. Their event is rarely the problem. **The right people never hear about it.** Reach is limited to friends and existing groups, so every new event starts from zero audience.

Building another registration portal doesn't fix this, because it gives people one more place to remember to visit. Participants already spend their time on WhatsApp, Instagram, LinkedIn, Discord and college communities. Team formation, reminders and follow-ups happen there too, but manually and in scattered ways.

The Challenge

Build an AI-powered system that **finds the right audience, reaches them on the platforms they already use, and lets them register without leaving those platforms**. It must work for a first-time organizer with no followers and no budget. Then show the world what you built through a short content video.

What We Need

A. The Solution (your solution must answer: *"How do the first 100 registrations come, if the organizer starts with nothing?"*)

1. **Audience discovery:** Identify the communities, clubs and colleges most likely to care about an event, and draft personalized outreach for each.
2. **In-chat registration:** Registration, FAQs and reminders handled entirely inside WhatsApp, Telegram or Instagram DMs, in Hindi, English and one regional language.
3. **Growth loops:** Features that make every participant bring more participants, such as "invite 2 teammates to complete your team" or shareable AI-generated team cards.
4. **Team matchmaking and drop-off prevention:** Match solo participants into balanced teams and nudge likely no-shows with personalized reminders.

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Rules

- Use **synthetic or sample data only**. No real participant data.
- **No spam.** All outreach must be opt-in and approved by a human before sending.
- AI-generated promotions must be **honest**: no fake numbers, testimonials or claims.
- Must be usable by a **non-technical organizer**.

Deliverables

- A working prototype with a live demo of one full journey: *event created → audience found → message sent → registered inside chat*
- A 1-slide **growth plan** showing how the first 100 users would be acquired and how it scales to 1,000

3.) **Deepfake Arena: Red Team vs Blue Team**

The Problem

AI-generated fake videos, voices and images are cheap, fast and convincing, and most people can't tell real from fake. Detection tools are rarely tested against a creative, determined attacker. Cybersecurity solved this with red teams and blue teams, but misinformation defense has no such game yet.

The Challenge

Every team builds **two things**:

- **Forger (Red):** creates convincing synthetic media, using only the provided synthetic personas, to fool the other teams.
- **Detective (Blue):** an AI system that tells real from fake and explains why.

How the Game Works

1. **Forge:** Each team creates 5 fake items (image, short video, audio or text).
2. **Mix:** Organizers combine all fakes with verified real items into one unlabeled Evidence Pool.
3. **Detect:** Each Detective gives a verdict, confidence level and reason for every item.
4. **Reveal:** Organizers reveal the truth live, item by item, and the scoreboard updates.

What We Need

- **Forger:** a pipeline to generate synthetic media from the provided personas.
- **Detective:** a system that outputs verdict, confidence and a human-readable reason (for example "lip movement doesn't match audio").
- **Explanation view:** highlight what triggered suspicion so a non-expert can understand.
- **Post-game report:** which fakes slipped through, why, and how to improve.
- **Content video:** a 45 to 90 second vertical (9:16) reel or ad with behind-the-scenes footage, the Red vs Blue rivalry, and a short demo, made using AI tools. One team member owns it.

Scoring

Your fake fools another team: **+10** per team. Detective catches a fake: **+5**. Correctly identifies a real item: **+3**. Correct reason: **+3**. False alarm on a real item: **−5**. Missed fake: **−3**.

Rules

- **Synthetic personas only.** No real people, politicians, celebrities, officials or classmates.
- All fakes stay inside the hackathon and are **never posted** online or on messaging apps.
- The content video must **label AI-generated parts** (for example "made with AI") and use no real person's likeness without consent.

Deliverables

Working Forger and Detective prototypes, your 5 submitted fakes and verdicts, a post-game report, the content video, and a 5-minute pitch.

Note: This problem statement will be confirmed **only if at least 2 teams select it**, since the game needs opposing teams. If fewer than 2 teams choose it, the teams that selected it must switch to another problem statement.

4.) AI for Girls' and Women's Safety

The Problem

For millions of girls and women, travelling to college, returning from work, using public transport, or walking alone after dark can create serious safety concerns. In an emergency, reaching the right person quickly can be difficult. A person may be unable to unlock their phone, make a call, explain their location, or manually trigger an SOS alert.

Existing safety applications often depend on users opening an app and pressing an emergency button. They may offer limited support when a person cannot interact with their device, is in an unfamiliar location, or needs immediate assistance. Meanwhile, trusted contacts may not know whether someone has reached their destination safely.

The real challenge is not just sending an SOS alert. It is identifying potential danger, reducing the time needed to seek help, and connecting the person at risk to the right support.

The Challenge

Build an AI-powered women's safety system that **helps detect potential danger, enables discreet emergency alerts, and coordinates timely assistance using a person's phone and available trusted support network.**

The solution should work in real-world situations, including late-night travel, isolated areas, public transport, and emergencies where the user cannot easily interact with their phone.

What We Need

A. The Solution (your solution must answer: *"How can a woman get timely help when she cannot safely make a call or manually trigger an SOS?"*)

1. **AI-Powered Risk Detection:** Use contextual signals such as unusual route deviations, prolonged unexpected stops, or user-triggered distress signals to identify situations that may require attention. Explain how your system distinguishes potential danger from ordinary behaviour and avoids excessive false alarms.
2. **Discreet SOS and Emergency Communication:** Create a quick or discreet way to request help through a wearable, phone shortcut, voice command, or another innovative interface. Share the user's location and an emergency alert with designated trusted contacts after a verified trigger.
3. **Smart Safety Network:** Build a system that connects users with trusted contacts, verified campus security personnel, or other appropriate support services. Provide clear escalation steps if an alert goes unanswered, without exposing a user's location to unauthorised people.
4. **Safe Journey and Prevention:** Allow users to share a journey with trusted contacts, set an expected arrival time, and receive check-in reminders. If a check-in is missed, offer a safe, configurable escalation process. AI may analyse journey context to suggest safer options, while clearly communicating uncertainty.
5. **Multilingual and Accessible Support:** Support Hindi, English, and one regional language. Design the experience for users with different levels of technical literacy, including a simple interface that can be used under stress.

B. The Content Video (one team member owns this)

Create a 45–90 second reel or advertisement that tells the story of your project. It should include:

- A relatable opening scenario showing a safety concern during a commute or late-night journey.

- A clear hook that states the problem within the first 5 seconds.
- Behind-the-scenes moments of the team building the prototype during the hackathon.
- A short demonstration of the complete safety workflow, from distress trigger to alert and location sharing.
- AI used in creating the video, such as AI voiceover, generated visuals, animation, auto-captions, or AI-assisted editing.

Rules

- Use **synthetic or sample data only**. Do not use real women's location histories, personal information, or emergency records.
- **Privacy first:** Location tracking must be consent-based, limited to the necessary period, and visible to the user.
- **No unauthorised surveillance:** Do not secretly track individuals, identify people without consent, or publicly expose live locations.
- **Human oversight:** AI risk scores must not be treated as proof of a crime or danger. Provide ways to cancel false alarms and verify alerts where feasible.
- **Honest AI:** Clearly communicate limitations. Do not claim that the system can guarantee safety or automatically contact emergency services unless that capability is genuinely implemented and tested.
- **Reliable emergency handling:** Include fallback options for poor connectivity, failed notifications, and unanswered alerts.
- The solution must be usable by a non-technical user.

Deliverables

1. **A working prototype** with a live demonstration of one complete journey: *journey started → potential risk or distress trigger → SOS alert generated → location shared with an authorised trusted contact → acknowledgement or escalation.*
2. **A 1-slide impact and deployment plan** explaining how the solution could be piloted in a college campus or local community, how users would onboard trusted contacts, and how it could scale to 1,000 users while protecting their privacy.
3. **A short content video** of 45–90 seconds demonstrating the problem, prototype, and intended social impact.

5.) **AI-Powered Gamified Learning: From Theory to Practice**

The Problem

Students today have access to countless online courses, lectures, PDFs, and learning platforms. Yet, many struggle to understand complex concepts, remember what they learn, and apply theoretical knowledge to real-world problems.

Students are overwhelmed with theory but lack engaging, hands-on ways to turn knowledge into practical skills. Traditional online learning often feels passive: watch a lecture, read notes, take a quiz, and move on. Students may score well in tests but still struggle to build projects, solve real-world challenges, or understand how their knowledge applies beyond the classroom.

The problem is not a lack of learning resources. It is the lack of an engaging learning experience that combines theory, practice, experimentation, and measurable skill development.

The Challenge

Build an AI-powered learning platform where **students learn by playing, experimenting, solving challenges, and building real-world projects instead of only watching lectures and memorising theory.**

The platform should transform any subject into an interactive learning journey where AI acts as a personalised mentor, games make learning engaging, and practical challenges help students develop skills they can actually use.

What We Need

A. The Solution (your solution must answer: *"How can a student go from understanding a concept to applying it in a real-world project through an engaging, game-like experience?"*)

1. **AI-Powered Personalised Learning:** Identify a student's knowledge gaps, learning pace, and preferred difficulty level. Generate personalised explanations, interactive exercises, and challenges that adapt as the student progresses.
2. **Learn by Playing:** Convert theoretical topics into missions, quests, simulations, puzzles, and competitive challenges. Students should earn points, unlock levels, complete achievements, and learn through experimentation rather than simply memorising answers.
3. **Theory-to-Practice Simulator:** Allow students to apply concepts in interactive environments. For example, programming students can debug code, AI students can train simple models, business students can manage virtual startups, and engineering students can solve simulated real-world problems.
4. **AI Mentor and Instant Feedback:** Provide a conversational AI mentor that explains why an answer is incorrect, offers hints instead of immediately revealing solutions, and guides students through difficult concepts in Hindi, English, and one regional language.
5. **Real-World Projects and Skill Proof:** Turn completed learning paths into practical mini-projects. Evaluate students on their problem-solving process, project outcomes, and understanding—not just quiz scores. Generate a portfolio showcasing demonstrated skills and completed challenges.
6. **Collaborative Learning:** Enable students to form teams, solve timed challenges, compete in learning tournaments, and collaborate on projects. Reward meaningful progress and teamwork rather than only speed or leaderboard position.

B. The Content Video (one team member owns this)

Create a 45–90 second reel or advertisement that tells the story of your project. It should include:

- A strong hook in the first 5 seconds: "What if learning felt more like playing a game than preparing for an exam?"
- A relatable scene showing a student struggling with theory and forgetting what they studied.
- Behind-the-scenes moments of the team building the prototype during the hackathon.
- A short demo of the platform, showing a concept becoming a mission, an interactive challenge, and a practical outcome.
- AI used in making the video, such as AI voiceover, generated visuals, animation, auto-captions, or AI-assisted editing.

Rules

- Use **synthetic or sample data only**. Do not use real student records or personal academic information.
- **AI must add real value:** Personalisation, adaptive challenges, feedback, or interactive simulations should be meaningfully powered by AI.
- **Learning over gaming:** Points and rewards must encourage actual understanding and skill development, not mindless engagement.

- **Honest evaluation:** Do not claim a student has mastered a skill solely because they completed a quiz or earned points.
- **Accessible by design:** The platform must be usable by students without advanced technical knowledge and should support different learning levels.
- **Working prototype required:** Teams should demonstrate at least one complete learning journey rather than only a UI mock-up.
- The solution must be usable by a non-technical student.

Deliverables

1. **A working prototype** with a live demo of one complete journey: *student selects a topic → AI assesses their level → an interactive game or mission begins → the student solves a practical challenge → AI provides feedback → a skill or project outcome is unlocked.*
2. **A 1-slide adoption and impact plan** showing how the platform would attract its first 100 students, encourage continued learning, and scale to 1,000 students. Include how learning outcomes will be measured beyond engagement metrics.
3. **A short content video** of 45–90 seconds demonstrating the learning experience, the working prototype, and its potential impact on students.

6.) AI for Smarter Healthcare

The Problem

Millions of people struggle to understand their symptoms, medical reports, and health information. They often rely on unreliable online sources or delay seeking medical help due to a lack of awareness and accessible guidance.

The challenge is to make healthcare information simple, accessible, and actionable using AI, without replacing medical professionals.

The Challenge

Build an AI-powered healthcare platform that helps users understand health concerns, learn about preventive care, and identify appropriate next steps through personalised guidance and interactive experiences.

What We Need

A. The Solution

1. **AI Health Assistant:** Explain common symptoms and medical terms in simple language, supporting Hindi, English, and one regional language.
2. **Smart Symptom Guidance:** Provide responsible, preliminary guidance on appropriate next steps, including when to seek medical attention.
3. **Medical Report Simplifier:** Explain synthetic lab reports, medical terminology, and reference ranges in an easy-to-understand format.
4. **Preventive Healthcare:** Offer personalised health education, medication reminders, appointment reminders, and wellness tracking.

B. The Content Video

Create a 45–90 second video featuring:

- A strong opening hook highlighting a real healthcare problem.
- Behind-the-scenes footage of your team building the prototype.
- A short demonstration of the AI healthcare solution.
- AI-generated voiceover, visuals, captions, or editing.

Rules

- Use synthetic or sample data only; no real patient records.
- AI must not replace doctors or claim to provide definitive diagnoses.
- Protect user privacy and communicate AI limitations.
- Include appropriate guidance for medical emergencies.
- The solution must be accessible to non-technical users.

Deliverables

1. **Working Prototype:** Demonstrate a complete journey from entering a health concern to receiving understandable guidance and appropriate next steps.
2. **One-Slide Impact Plan:** Explain how to acquire the first 100 users, scale to 1,000, and measure impact.
3. **Content Video:** A 45–90 second demonstration of the problem, solution, and its potential healthcare impact.